Tomáš Gonda

Researcher in Applied Category Theory and Quantum Information Theory

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Education

PhD **Perimeter Institute & University of Waterloo**, Physics, Quantum Information — Waterloo, ON, Canada, Sept 2017 – Nov 2021
- Thesis: *Resource theories as quantale modules*
- Supervisor: Dr. Robert W. Spekkens

MSc **Perimeter Institute & University of Waterloo**, Theoretical Physics — Waterloo, ON, Canada, Aug 2016 – June 2017
- Thesis: *Resource theoretic approach to causal inference*
- Supervisor: Dr. Robert W. Spekkens

BSc **University of Edinburgh**, Mathematics & Physics — Edinburgh, Scotland, Oct 2013 – May 2016
- First class honours, 92% average grade
- Thesis: *Symplectic Geometry*
- Supervisor: Dr. Johan Martens

Experience

University of Innsbruck, Senior postdoctoral researcher — Innsbruck, Austria, Mar 2025 – present, 1 year 1 month
Independent research in applied category theory and quantum information.
- Algebra group, led by Tim Netzer
- Funded by my FWF ESPRIT Grant Ergodic Theory in Categorical Probability

University of Innsbruck, Postdoctoral researcher — Innsbruck, Austria, Jan 2022 – Feb 2025, 3 years 2 months
Conducting research in applied category theory and quantum information theory.
- Mathematical quantum physics group, led by Gemma De Les Coves

Applied Category Theory Adjoint School, Supervisor / teaching assistant — Oxfork, UK, June 2024
Co-leading a summer school research project with Paolo Perrone.
- Topic: *Modelling Uncertainty with Markov Categories*

University of Waterloo & University of Innsbruck, Supervisor / project lead — varied, 2022 – present, 4 years
Supervising Master's thesis projects of:
- Yìlè Yīng (MSc, 2022, co-supervised by RW Spekkens); *Asymmetry Trade-off Relations: A New Kind of Uncertainty Relations*
- Stefan Kremminger (MSc, 2024); *Quantum Conditional Independence*

University of Waterloo, Teaching assistant — Waterloo, ON, Canada, 2017 – 2021, 4 years
- Conducting tutorials for graduate level courses in Quantum Theory and Statistical Physics.
- Supervising multiple Master's level winter school projects in Quantum Foundations.

ETH Zürich, Visiting researcher — Zürich, Switzerland, Summer 2019
Collaborating with researchers in the group of Renato Renner on topics in information theory.

InstaEDU, Online tutor — remote, 2013 – 2016, 3 years
Online tutoring of undergraduate students in science and engineering.

University of Edinburgh, Career development internship — Edinburgh, Scotland
A summer project on modelling turbulence in Matlab. — Summer 2015
- Topic: *Transition to turbulence in plane Couette flow*

Personal tutor — Bratislava, Slovakia
In-person tutoring of high school students. — 2012 – 2013
1 year

Volunteering

Scientific Journals, Reviewer — remote
Reviewing manuscripts for:
- Philosophy of Science (1)
- Quantum (3)
- Compositionality (2)
- Journal of Pure and Applied Algebra (1)
- Journal of Mathematical Physics (1)
- various conference submissions.

Positive Futures Festival, Organizer — Innsbruck, Austria
Co-organizing an annual two week long music festival with ~40 concerts and ~1000 attendees. — 2024 – present

Touch&Play Cologne Festival, Organizer — Germany
Co-organizing and facilitating an annual week long retreat with ~100 attendees. — 2025 – present

Awards

ESPRIT Grant Ergodic Theory in Categorical Probability — Mar 2025
The ESPRIT Grant supports young independent researchers in Austria.
Austrian Science Fund
fwf.ac.at/forschungsradar/10.55776/ESP3451824

Napier Medal and Gangadhar Balwant Gadgil Prize — 2016
Awarded to the best student in the Year 4 of the Mathematics program.
University of Edinburgh

Nichol Foundation Scholarship — 2015
Awarded to the best overall mark in the Year 3 of Physics degrees.
University of Edinburgh

Kelland Memorial Prize — 2014
Awarded for distinguished performance for direct entrants into the Second year of the program.
University of Edinburgh

Selected Publications

Empirical measures and strong laws of large numbers in categorical probability — Mar 2025
We introduce axioms for empirical sampling morphisms, which are abstract notions of empirical measures within the framework of categorical probability. Using these, we prove abstract versions of the de Finetti theorem, the Glivenko–Cantelli theorem, and the strong law of large numbers, and show that classical results are recovered via the category of partial Markov kernels.
Tobias Fritz, **Tomáš Gonda**, Antonio Lorenzin, Paolo Perrone, Areeb Shah Mohammed
arxiv.org/abs/2503.21576 (preprint)

De Finetti's theorem in categorical probability Sept 2021

A novel proof of de Finetti's Theorem characterizing permutation-invariant probability measures of infinite sequences of variables. The proof is phrased in the language of Markov categories, which provide an abstract categorical framework for probability and information flow. The diagrammatic and modular nature of the arguments makes the proof intuitive and easy to follow.
Tobias Fritz, **Tomáš Gonda**, Paolo Perrone
repository.lsu.edu/josa/vol2/iss4/6 (Journal of Stochastic Analysis)

Epistemic horizons from deterministic laws: Lessons from a nomic toy theory Mar 2025

A study of what embedded observers can learn about the rest of the world made up of classical particles. We show that, in spite of deterministic physical laws, they face an epistemic horizon — properties such as position and momentum cannot be known simultaneously.
Johannes Fankhauser, **Tomáš Gonda**, Gemma De Les Coves
link.springer.com/article/10.1007/s11229-024-04852-0 (Synthese)

A framework for universality in physics, computer science, and beyond Aug 2024

First steps towards an abstract theory of universality, generalizing universal Turing machines, universal spin systems, completeness for complexity classes,and much more.
Tomáš Gonda, Tobias Reinhart, Sebastian Stengele, Gemma De Les Coves
compositionality.episciences.org/14134 (Compositionality)

Monotones in general resource theories Aug 2023

Resource theories offer an abstract framework for comparing resources for all kinds of tasks. We provide a general theory for quantifying the cost and yield of a resource, generalizing a plethora of known constructions.
Tomáš Gonda, Robert W. Spekkens
compositionality.episciences.org/13526 (Compositionality)

Selected Presentations

Introduction to categorical probability Waterloo, ON, Canada
A 4-lecture mini-course at the Perimeter Institute Oct 2025
- Online recordings: pirsa.org/c25054

Introduction to categorical probability Vienna, Austria
IQOQI Vienna Seminar Sept 2025
- Youtube recording: youtube.com/watch?v=eVfFuIGEZxc

Resource dependence relations and contextuality in asymmetry trade-offs Varna, Bulgaria
Quantum Physics and Logic (QPL) Conference July 2024

Epistemic horizons from deterministic laws: Lessons from a nomic toy theory Oxford, UK
OASIS — The Oxford Advanced Seminar on Informatic Structures June 2024
- Invited talk

A framework for universality in physics, computer science, and beyond remote
NYC category theory seminar Sept 2023
- Invited talk
- Youtube recording: youtube.com/watch?v=8_ZX2vpqBZU

Supports and idempotent splitting in Markov categories Maryland, USA
Applied Category Theory Conference Aug 2023
- Youtube recording: youtube.com/watch?v=3QJzfdZu8_g

A framework for universality in physics, computer science, and beyond — Bergen, Norway
Category Theory at Work in Computational Mathematics and Theoretical Informatics (CATMI) — June 2023
- Invited talk

De Finetti's theorem in categorical probability — Cambridge, UK
Applied Category Theory Conference — July 2021

Blackwell-Sherman-Stein Theorem in categorical probability — remote
Applied Category Theory Conference — July 2020
- keynote presentation
- Youtube recording: youtube.com/watch?v=fgWUV-hE0CI

Almost quantum correlations are inconsistent with Specker's principle — Boulder, CO, USA
Quantum Information Processing (QIP) Conference — Jan 2019

Almost quantum correlations are inconsistent with Specker's principle — Florianopolis, Brazil
Q-Turn Conference — Nov 2018
- Youtube recording: youtube.com/watch?v=u2XwLiktTI8

The ontological nature of causality — Cambridge, UK
Rethinking Foundations of Physics Workshop — July 2018
- Youtube recording: youtube.com/watch?v=3QJzfdZu8_g

Peculiarities of quantum theory — Bratislava, Slovakia
FKS Summer School — Aug 2017

Languages

Slovak
Native

Czech
Fluent

English
Fluent (C1/C2)

German
Fluent (B2)